Ahan Bhatt

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Summary

M.S. in Computer Science and Engineering with 5 years of combined industry and research experience in AI and software engineering. Built and deployed production AI systems for enterprise clients at Intellytics and Growexx, spanning RAG, document intelligence, fraud detection, and healthcare automation. Developed transformer compilers, GPU-accelerated LLM inference, and adversarial evaluation systems. Skilled in Python, C++, CUDA, Triton, PyTorch, FastAPI, and scalable ML infrastructure, with published AI research.

Experience

(6 years total / 4.5 years full time)

Research Assistant — September 2025 – June 2026
AI Explainability and Accountability (AIEA) Lab @ UCSC [**https://aiea-lab.github.io/**] — *Santa Cruz, California*

- Built ReliaGuard Studio to audit harmful human-AI reliance using neuro-symbolic risk models and conformal gating.
- Developed BreakPoint to turn observed LLM failures into adversarial regression tests with multi-judge evaluation and CI gates.
- Engineered FastAPI/Next.js dashboards, PostgreSQL data pipelines and human-review workflows for explainable AI.

Artificial Intelligence Engineer — August 2024 – August 2025
Growexx [**https://www.growexx.com/**] — *Ahmedabad, Gujarat, India*

- Deployed a prior-authorization AI agent for a hospital and pharmacy client using FHIR, hybrid RAG, LangGraph and MCP.
- Recorded 92.5% first-pass acceptance, 84% automation coverage and 6.5 minutes of staff time per authorization.
- Delivered production multi-agent workflows for a major apparel retailer across CRM, inventory, payments and support.
- Automated returns, refunds and order resolution using Temporal workflows, tool guardrails and human escalation.

AI Researcher (Part-time) — June 2024 – May 2025
AI Institute of South Carolina [**https://aiisc.ai/**] — *Columbia, South Carolina (remote)*

- Researched multimodal safety with vision-language models and preference optimization for safer text-to-image generation.
- Benchmarked time-series foundation models against statistical and deep-learning baselines across five datasets.

Data Scientist — January 2024 – May 2024
Space Applications Centre, Indian Space Research Organization [**https://www.sac.gov.in/**] — *Ahmedabad, Gujarat, India*

- Deployed an LSTM satellite clock-bias forecasting platform with Kafka streaming and automated anomaly filtering.
- Achieved 2.11×10^{-11} RMSE on Galileo data and built a Next.js/React interface for live forecast monitoring.

Software Engineer (AI) — February 2022 – January 2024
Intellytics [**https://intellytics.com/**] — *Ahmedabad, Gujarat, Indiat*

- Built permission-aware RAG search for a 1,000+ employee consulting client across SharePoint, Snowflake and APIs.
- Developed a GraphRAG fraud investigation platform for a fintech client using Neo4j, Kafka, Snowflake and FastAPI.
- Built insurance document intelligence pipelines with OCR, structured extraction, hybrid search and human review.
- Deployed auditable Python/FastAPI services and data workflows with evidence citations, access controls and monitoring.

Education

University of California, Santa Cruz [3.82 GPA] — Santa Cruz, California
Master of Science in Computer Science and Engineering — *Aug. 2025 – Jun 2026*

Indus Institute of Technology and Engineering [3.94 GPA] — Ahmedabad, India
Bachelor of Technology in Computer Engineering — *Aug. 2020 – July 2024*

Recent Projects

CacheIR [**https://github.com/AhanBhatt/CacheIR**] [**CUDA, Triton, C++20, custom compiler IRs, PyTorch, CMake, pybind11, FastAPI, NumPy**]

- A narrow, inspectable compiler/runtime for GPU-accelerated decoder-only transformer inference, up to 3.24× faster than vLLM on certain workloads. It imports Llama/Mistral/Qwen models into a custom IR, applies transformer-specific compiler optimizations, plans memory/execution, selects CUDA kernels, and serves through a reference runtime.

BreakPoint [**https://github.com/AhanBhatt/BreakPoint**] [**LLM evaluation pipelines, multi-judge validation, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, MLflow, DuckDB**]

- A failure-to-eval compiler for LLM systems that mines real failure modes into versioned adversarial regression suites with DSL-defined traps/rubrics, multi-judge calibration, RAG/tool simulators, CI gates, FastAPI/Next.js dashboards, and 360 validated eval cases across 9 failure families.

Publications

https://scholar.google.com/citations?user=gteC3uwAAAAJ&hl=en

- **Leveraging LSTM for Predictive Modeling of Satellite Clock Bias**
  [**https://ieeexplore.ieee.org/document/10908787**] *Presented at CDMA 2025 conference and available on IEEE Xplore*
- **Generating Knowledge Graphs from Large Language Models** [**https://arxiv.org/abs/2412.07412**]
- **Time Series Foundational Models: Their Role in Anomaly Detection and Prediction**
  [**https://arxiv.org/abs/2412.19286**] *Presented at AAAI 2025 Anomaly Detection in Scientific Domains Workshop*

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R
Machine Learning: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, NumPy, Pandas
ML Systems and Compilers: CUDA, Triton, C++20, pybind11, custom IRs, operator fusion, High-Performance Computing (HPC), kernel selection, memory/execution planning, paged KV caching, quantization
LLM and Data Tools: LangChain, LangGraph, MCP, RAG, Hybrid RAG, ChromaDB, vector databases, Multi-Agent Systems, Agentic AI, Tool Calling, LoRA/QLoRA, LLM Evaluation, LLM Observability, AI Guardrails, Human-in-the-Loop (HITL)
Backend and Deployment: Node.js, FastAPI, REST APIs, FHIR, Docker, CMake, Bazel, Kubernetes, Linux, Git, CI/CD, Nginx
Cloud: AWS EC2, S3, IAM, Lambda, SageMaker, CloudWatch
Data Engineering: PostgreSQL, Redis, Real-Time Data Pipelines, ETL, schema design